3D modelling of confined gastruloid aggregates

Contents

1 The gastruloid physical model 1

2 Description of the experiment 1

3 Simulations 2

4 Adaptability and possible improvements 5

5 Parameters 7

1 The gastruloid physical model

Definition 1. *The **equations of motion** are given by*

$$\mathbf{v}_i = \frac{1}{\lambda n_i}\sum_j \mathbf{F}_{ij} + \frac{\alpha}{n_i}\sum_j \mathbf{v}_j\,.$$

***Cellular forces** are given by*

$$\mathbf{F}_{ij}(s_i, s_j, d_{ij}) = f(d_{ij})\,\hat{\mathbf{r}}_{ij}\cdot\begin{cases} F_{rep}^{(s_i,s_j)}, & \text{if } d_{ij} \le 2r \\ F_{att}^{(s_i,s_j)}, & \text{if } 2r < d_{ij} < 2\mu r \\ 0, & \text{otherwise}\,, \end{cases}$$

where $s_i, s_j \in \{A, B, C\}$, $d_{ij} = |\mathbf{x}_i - \mathbf{x}_j|$, $\hat{\mathbf{r}}_{ij} = \frac{\mathbf{x}_i - \mathbf{x}_j}{|\mathbf{x}_i - \mathbf{x}_j|}$, *and*

$$f(d_{ij}) = \left(\frac{2r}{d_{ij}} - 1\right)\left(\frac{2\mu r}{d_{ij}} - 1\right)\,.$$

Remark 1. *In the code we set* $\hat{\alpha} = \lambda\alpha$ *and so*

$$\mathbf{v}_i = \frac{1}{\lambda n_i}\left(\sum_j \mathbf{F}_{ij} + \alpha\sum_j \mathbf{v}_j\right)\,.$$

Definition 2. *The **division time** is chosen uniformly at random from*

$$U(\tau_{div}(1-\sigma_{div}), \tau_{div}(1+\sigma_{div}))\,.$$

Definition 3. *The **transition rates** depend on the proportion of cells in state A, namely*

$$\begin{aligned} r_{A\to B}^{(i)} &= \frac{p}{1+\phi_A/K} \\ r_{B\to C}^{(i)} &= \frac{q}{1+\phi_A/K}\,. \end{aligned} \tag{1}$$

2 Description of the experiment

In the original experiment as follows.

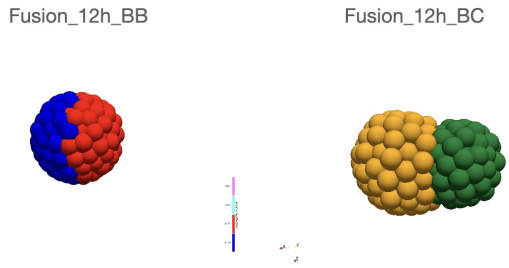


Figure 1: Fusion simulations carried out by Sotiris which replicate the experimental BB and BC fusions.

First, we grow and differentiate a gastruloid aggregate before polarisation begins. If we use the three-state cell description used in Oriola et al. 2026 ($A \to B \to C$), the resulting aggregate at this step is a spheroid with an outer layer of C cells and a central cluster of A, B cells. If we were to let this further evolve, this centroid would move away from the center to the outer layer (polsarisation).

Before this polarisation occurs, we confine this spherical aggregate inside two plates and observe different phenomena. On confinement, the result is a pancake-shaped aggregate inheriting the previous structue (an outer layer of C cells and a central cluster of A, B cells) but, after some time, this structure can evolve into two different phenomena. If the radius of the *pancake* is not wide enough, this cluster remains in the center and all its cells eventually differentiate to C. However, if the radius is large enough, then we observe some pattern-like structures formed by several AB-clusters that spread across the whole aggregate.

We note that the same number of cells that yields patterns for a given confinement thickness may not give rise to patterns for thicker *pancakes*, since their radius will be smaller.

3 Simulations

The main challenge of the simulations is to show the observed patterns for several confinement thicknesses and, using the same parameters, to replicate the following phenomena observed in the nonconfined gastruloid aggregates:

1. Polarisation of the central cluster.
2. Different behaviours for fusion of certain types of cells.

In order to fit our original parameters to the experiment, we start off using the parameters exposed in Table 5, which have been checked by Sotiris Samatas to yield the appropiate fusion results (see Figure 1).

Next, we tweak this parameter set slightly in order to attain both unconfined polarisation (see Figure 2) and confined patterns f(see Figure 3).

We also make sure that it works for any thickness and that we get both behaviours: a stable cluster that differentiates away to C if the number of cells N is not large enough at the moment of the confinement, and BC patterns otherwise. We define N_0, L the threshold N such that, for any $N > N_0, L$, a confined aggregate of thickness L will yield patterns. See both behaviours for two layers and three layers in Figures 4 and 5, respectively.

Finally, we sketch a phase space relating the appearance of patterns, the number of layers, and the initial $N = N_0$ at the moment of squeezing, shown in Figure 6

While tweaking the parameters, I have also observed stable behaviour that yield more concentrated patterns (much more pattern-like to the bare eye I think, but maybe not so true to the

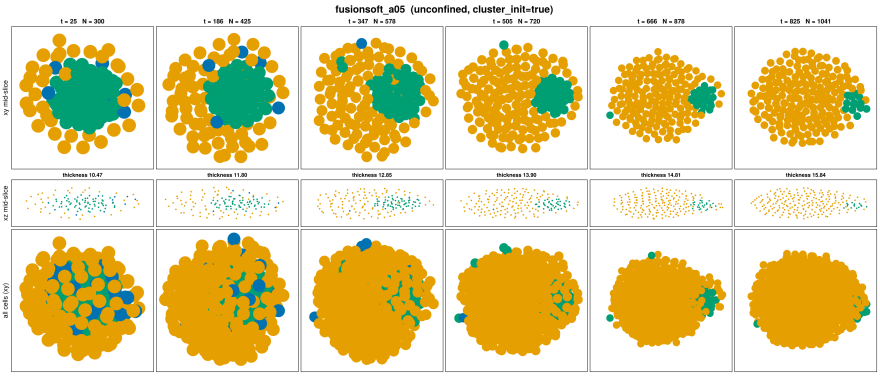


Figure 2: Unconfined polarisation of a 3D spherical aggregate using a 2D view. This could be plotted in 3D with no difficulty.

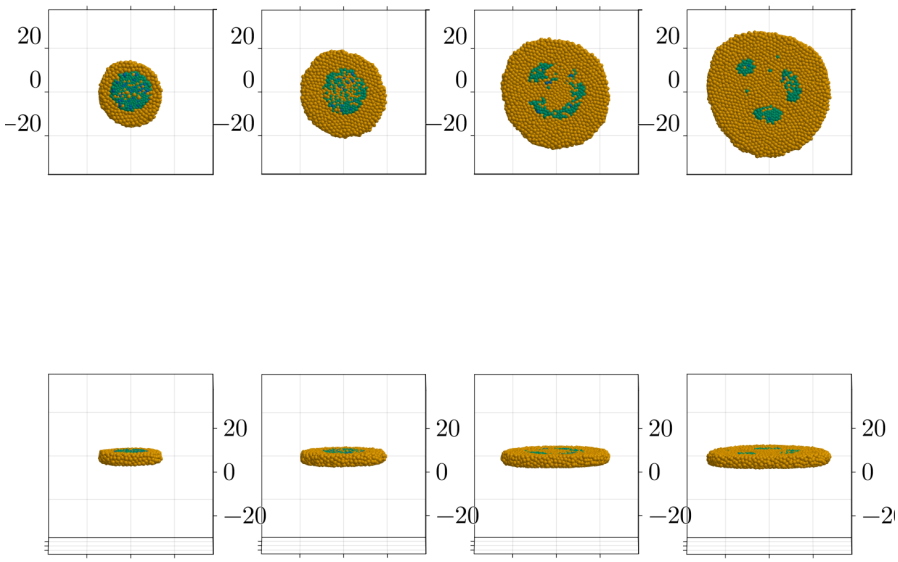


Figure 3: Example of patterns in a three-layer confined aggregate (that is, thickness= $3 * (2r)$).

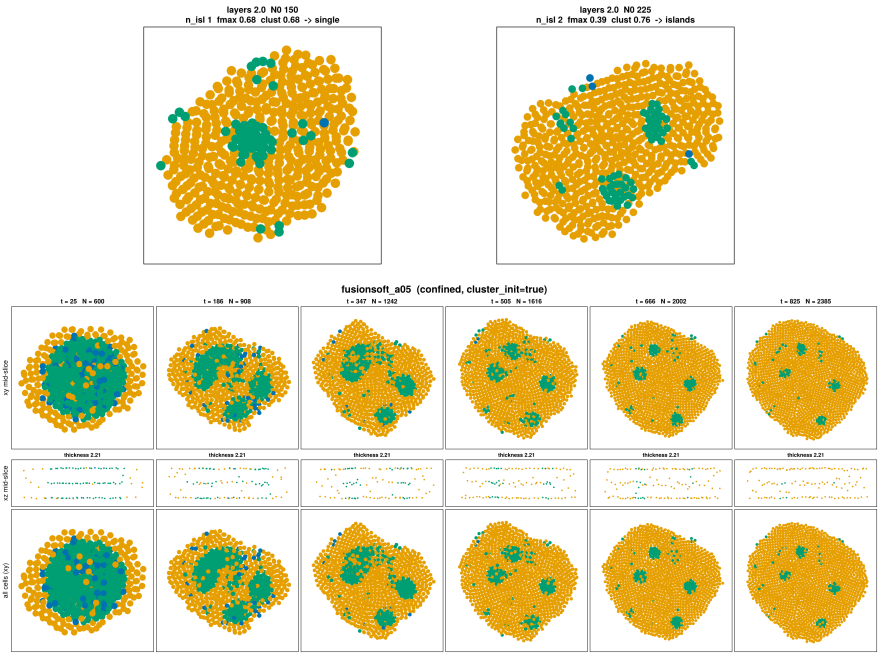


Figure 4: Two-layered aggregate with no patterns due to low N and with patterns with a larger N.

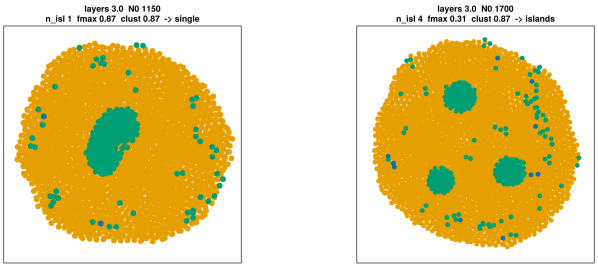


Figure 5: Two-layered aggregate with no patterns due to low N and with patterns with a larger N.

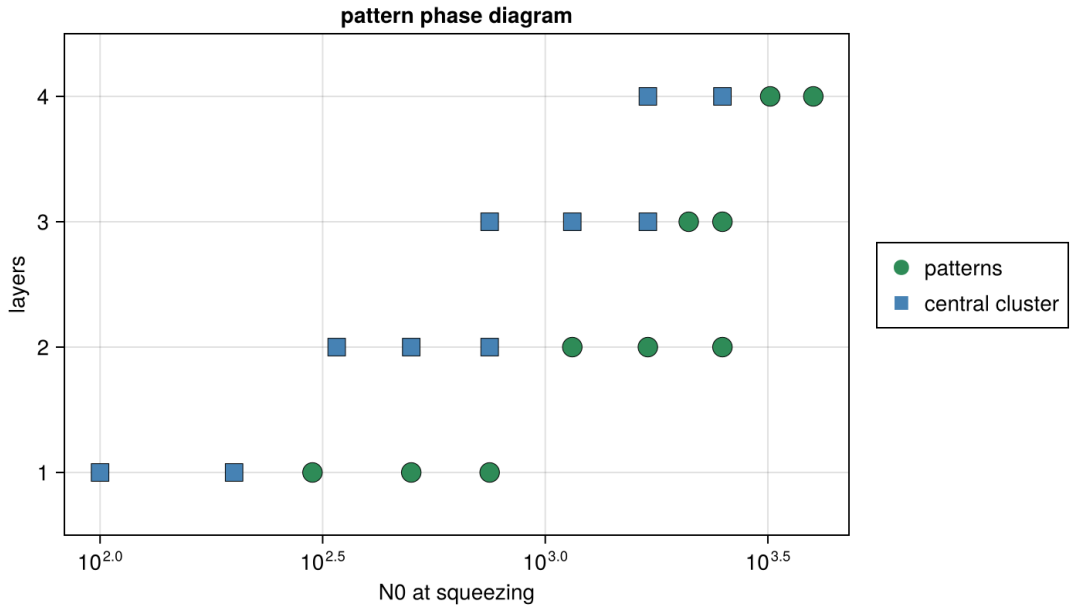


Figure 6: Two-layered aggregate with no patterns due to low N and with patterns with a larger N.

experiments). These are attached in Figure 7.

4 Adaptability and possible improvements

I believe this should be adaptable to new rates of change p, q relatively easily, since the algorithms for studying the correct functioning of the gastruloids (that is, polarisation, fusion, and patterning) are already established. Another possible change that we have discussed is considering local inhibition rates, which as far as I tested yield very similar results; at least we could try if it's preferred to have them that way.

The phase space plot could also be improved if you'd like to use it. I have two improvements in mind:

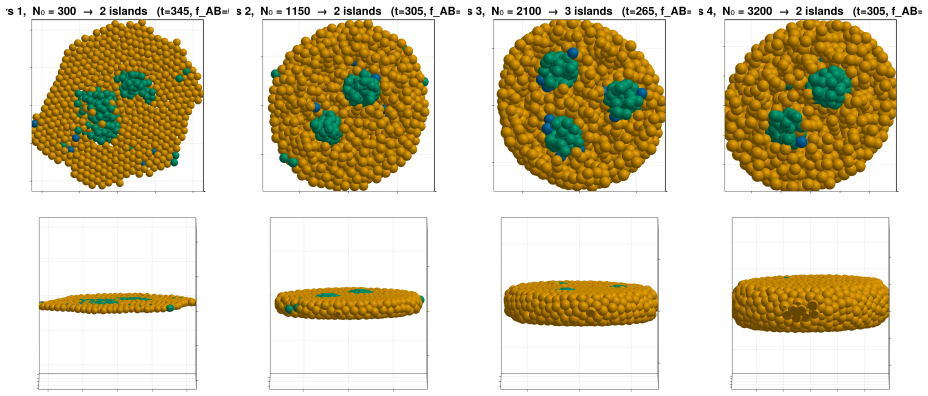


Figure 7: More concentred patterns for layers 1 to 4. Plot looks slightly uglier due to the engine used, and can be redone.

1. Plot it in a more continuous fashion: instrad of considering layers 1,2,3,4,5, let it have values in between. I believe we could fit the transition curve and paint either side blue / green depending on the behaviour.
2. Plot it in terms of the original radius on squeezing instead/apart from the initial N.
3. Be clearer about the classification: "if we run 5 aggregates with the same parameters and get patterns 3 out of 5 times, do we classify it as patterns?".

Symbol	Description	Value (code)	Value (exp)
Δt	Timestep	0.02	
r	Radius	1	
	Overlap factor	0.3	
α	Cell-ECM/cell-cell friction parameter	0.9	
τ_{div}^{A}	Average division time	300	
τ_{div}^{B}	Average division time	300	
τ_{div}^{C}	Average division time	750	
σ_{div}	Division time variability	0.5	
λ	Global friction coefficient	1	
μ	Force range factor	$\begin{pmatrix} 1.55 & 1.5 & 1.525 \\ 1.5 & 1.7 & 1.45 \\ 2.0 & 2.0 & 2.0 \end{pmatrix}$	
	Neighbour range factor	1	
F_{rep}	Repulsion force	$\begin{pmatrix} 3 & 3 & 3 \\ & 3 & 3 \\ & & 5.1 \end{pmatrix}$	
F_{att}	Attraction force	$\begin{pmatrix} 6 & 6 & 2.4 \\ & 6 & 1.6 \\ & & 4.8 \end{pmatrix}$	
p	Fixed transition rate from A to B	0.009333	
q	Fixed transition rate from B to C	0.004667	
λ	Friction parameter	1	
K	Feedback parameter	0.2	
n_i	Number of neighbours		
n_i^X	Number of neighbours in state X		
ϕ_X	Proportion of cells in state X		
T_0	Typical time scale	2 h	
R_0	Typical length scale	5 µm	
F_0	Typical force scale	20 pN	

Table 2: Base parameters used for the simulations.

5 Parameters